

SH-wave propagation from a point source in a medium with two layers

Chapter 6

On October 8, 1909, a massive earthquake occurred in the Kulpa Valley, causing severe damage. A Croatian meteorologist and seismologist, Andrija Mohorovičić, collected seismograms across Europe and plotted the travel-time curve, as shown in Figure 6.1. Based on these observations, he discovered a structural discontinuity at a depth of 54 km. He estimated the P-wave velocity of the crust to be 5.68 km/s, whereas that of the underlying mantle was 7.75 km/s. This boundary is now known as the Mohorovičić discontinuity, or Moho discontinuity for short.⁽²⁾ In this chapter, we will learn how to infer such discontinuities based on the behavior of reflected and refracted waves originating from a point source. For simplicity, we focus on SH wave propagations.

§6.1 Plane wave decomposition

In the previous chapter, we learned the reflection and refraction of plane waves. They are fundamental to interpreting seismic wave propagation because a seismic wave field can be represented by a superposition of plane waves as,

$$\phi(\mathbf{x}, t) = \int_{-\infty}^{\infty} d\omega \int_{-\infty}^{\infty} dk_x dk_y A(k_x, k_y, \omega) e^{i \left[k_x x + k_y y + \left(\frac{\omega^2}{\alpha^2} - k_x^2 - k_y^2 \right)^{1/2} z - \omega t \right]}. \quad (6.1)$$

Based on these results, we will consider seismic wave propagation in a two-layered medium, alongside the features of Green's functions in an infinite medium. For simplicity, we will focus on SH waves (which are mathematically equivalent to acoustic waves) in this chapter.

Let us review the Weyl integral representation explained in Section 5.6.

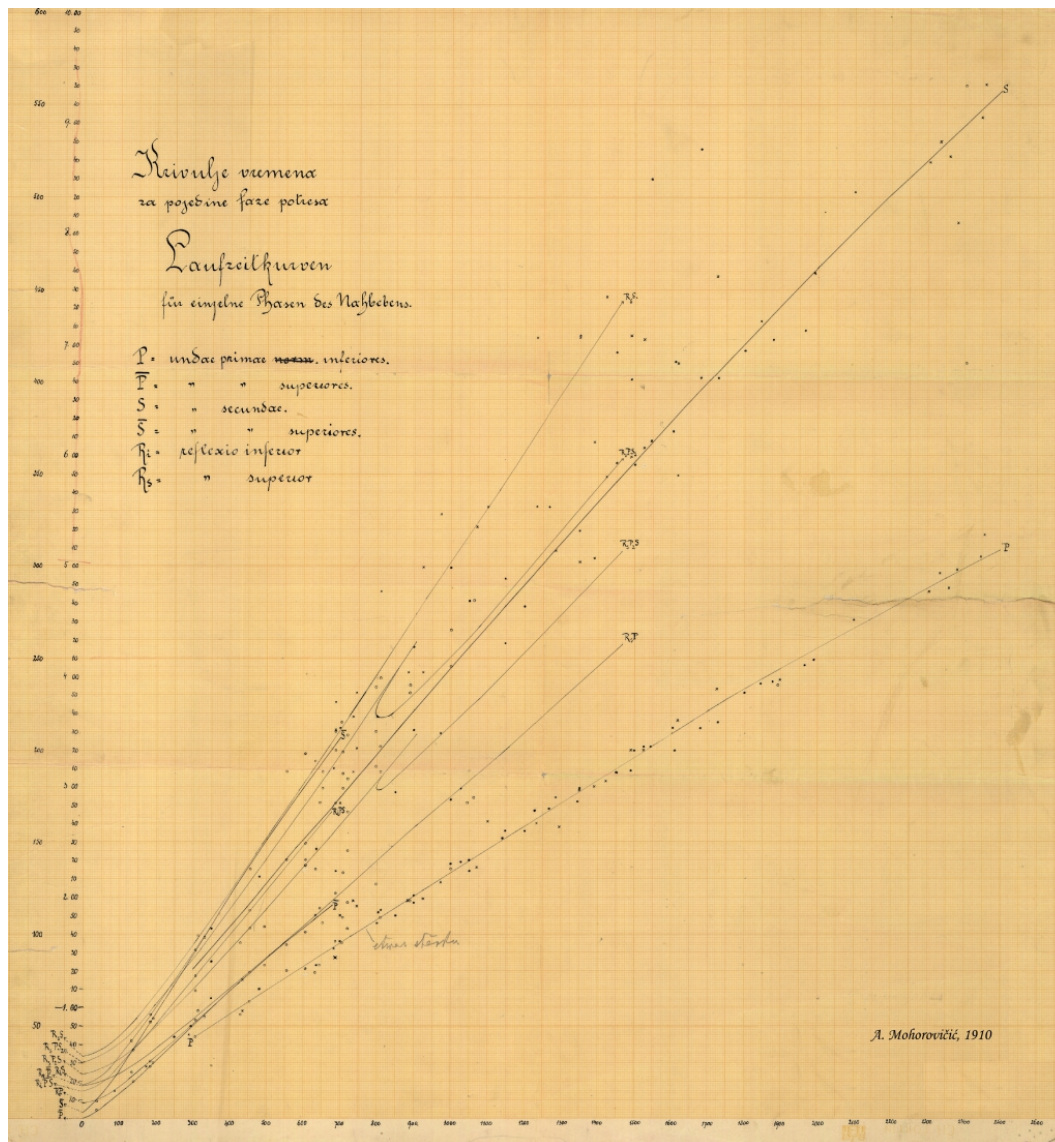


Figure 6.1: Observed travel-time curves for the 1909 earthquake, taken from a paper translated into English (4) from Croatian⁽⁴⁾ by Mohorovičić (1910).

§6.2 Reflection and refraction on an internal boundary

6.2.1 SH wave

Next, we consider reflection and refraction on a boundary inside an infinite elastic medium (Figure 6.2). A semi-infinite medium 2 $z < 0$ is welded to the medium 1 $z > 0$. The incident SH wave enters downward in medium 1, then it is reflected in medium 1 and refracted to medium 2. Corresponding plane wave solutions are given by

$$z > 0 : s_y = A_1 e^{-i\omega(t-p_{1x}x-p_{1z}z)} + B_1 e^{-i\omega(t-p_{1x}x+p_{1z}z)} \quad (6.2)$$

$$z < 0 : s_y = A_2 e^{-i\omega(t-p_{2x}x-p_{2z}z)}. \quad (6.3)$$

The first term of s_y for $z > 0$ represents the incident wave, the second term represents the reflected wave, and s_y for $z < 0$ represents the refracted (transmitted) wave in medium

2. To satisfy the boundary condition for the continuity of displacement, the horizontal slowness must be continuous across the boundary, meaning $p_{1x} = p_{2x}$. This condition is mathematically identical to Snell's law:

$$\frac{\sin \theta_1}{\beta_1} = \frac{\sin \theta_2}{\beta_2}. \quad (6.4)$$

To satisfy the condition of continuity of displacement at $z = 0$, the relation $A_1 + B_1 = A_2$ is required. Furthermore, to satisfy the boundary condition for the continuity of stress, $\mu_1 p_{1z}(A_1 - B_1) = \mu_2 p_{2z} A_2$ is also required. Consequently, the corresponding reflection coefficient R_{12} and transmission coefficient T_{12} are given by:

$$R_{12} = \frac{B_1}{A_1} = \frac{\mu_1 p_{1z} - \mu_2 p_{2z}}{\mu_1 p_{1z} + \mu_2 p_{2z}} = \frac{\rho_1 \beta_1 \cos \theta_1 - \rho_2 \beta_2 \cos \theta_2}{\rho_1 \beta_1 \cos \theta_1 + \rho_2 \beta_2 \cos \theta_2} \quad (6.5)$$

$$T_{12} = \frac{A_2}{A_1} = \frac{2\mu_1 p_{1z}}{\mu_1 p_{1z} + \mu_2 p_{2z}} = \frac{2\rho_1 \beta_1 \cos \theta_1}{\rho_1 \beta_1 \cos \theta_1 + \rho_2 \beta_2 \cos \theta_2}. \quad (6.6)$$

The continuity of displacement at the boundary leads to the relation $1 + R_{12} = T_{12}$. You can explore a playful web application demonstrating reflection and transmission phenomena at: https://www.eri.u-tokyo.ac.jp/people/knishida/eng/Seismology/wave_coe.html.

Conservation of energy

The total energy flux entering a unit area at $z = 0$, given by $K_{in} \cos \theta_1$, must equal the sum of the outgoing energy fluxes from the reflected wave ($K_{refl} \cos \theta_1$) and the refracted wave ($K_{trans} \cos \theta_2$).

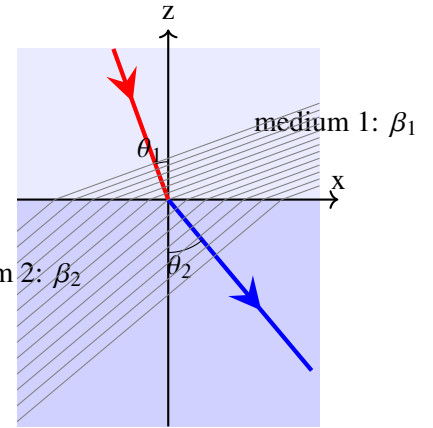


Figure 6.2:

In the case of SH waves, the conservation of energy is expressed as:

$$\rho_1 \beta_1 \cos \theta_1 = \rho_1 \beta_1 \cos \theta_1 |R_{12}|^2 + \rho_2 \beta_2 \cos \theta_2 |T_{12}|^2. \quad (6.7)$$

Total reflection

When $\theta_2 > \theta_1$, total reflection occurs for an incident angle θ_1 that is larger than the critical angle θ_c , which is defined by:

$$\theta_c = \sin^{-1} \frac{\beta_1}{\beta_2}. \quad (6.8)$$

In this case, since p_{2z} becomes purely imaginary, the amplitude of the refracted wave in medium 2 decreases exponentially with depth z . Consequently, the refracted wave does not transfer energy downward into medium 2.

On the other hand, although the reflection coefficient R_{12} is a complex number, its absolute value is exactly 1 (since the numerator is the complex conjugate of the denominator). Considering the conservation of energy in Eq. 6.7, $|R_{12}|^2 = 1$ and $|T_{12}|^2 = 0$. This also indicates that the phase of the reflected wave is advanced during total reflection. The phase shift of R_{12} is given by:

$$\arg R_{12} = -2 \tan^{-1} \frac{\mu_2 i p_{2z}}{\mu_1 p_{1z}} = -2 \tan^{-1} \frac{\mu_2 \sqrt{p^2 - \beta_2^{-2}}}{\mu_1 \sqrt{\beta_1^{-2} - p^2}} \quad (6.9)$$

When μ_2 goes infinity, the reflection coefficient $R_{12} = -1$ represents reflection at a rigid wall.

Acoustic impedance

For simplicity, here we consider Eq. ?? for vertically propagating wave ($\theta = 0$). The reflection and transmission coefficients are give by

$$R_{12} = \frac{\rho_1 \beta_1 - \rho_2 \beta_2}{\rho_1 \beta_1 + \rho_2 \beta_2} \quad (6.10)$$

$$T_{12} = \frac{2\rho_1 \beta_1}{\rho_1 \beta_1 + \rho_2 \beta_2}. \quad (6.11)$$

$\rho\beta$ is called acoustic impedance, which shows resistance of entering wave. The equation shows the impedance contrast determines the reflection and transmission coefficients. The acoustic impedance is defined as the ratio between stress and particle velocity as an analogy of resistance of the electric circuit.

With increasing $\rho_2 \beta_2$, the transmitted wave is harder to enter. When the two acoustic impedances match with each other as $\rho_1 \beta_1 = \rho_2 \beta_2$, no reflection occurs. Even if seismic velocities of two media are the same, a reflected wave occurs owing to the density contrast. In summary, the reflection and transmission coefficients do not contain information on seismic velocity contrast but also on density contrast. The coefficients are crucial for exploring density contrast at discontinuities of the Earth.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

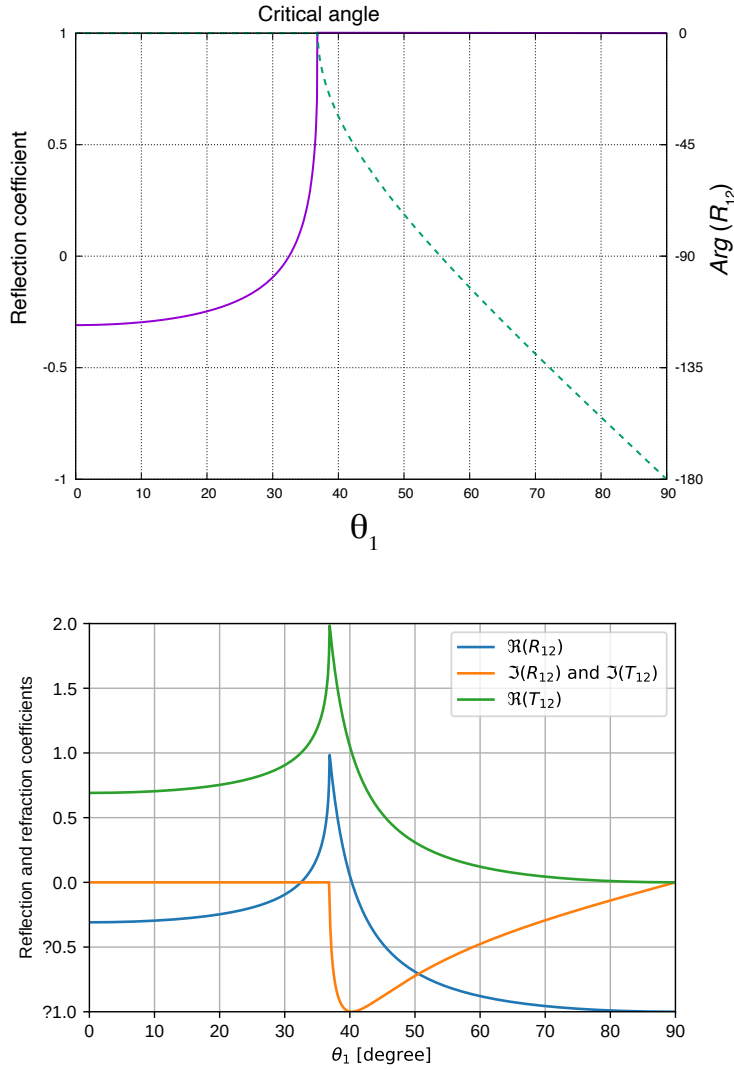


Figure 6.3: (Upper) Reflection coefficient R_{12} of SH wave on the internal boundary. The coefficient is real before the critical angle, and it becomes complex after the angle. The solid line shows the reflection coefficient, and the broken line shows the phase. (Lower) Reflection and Refraction coefficients R_{12} and T_1 . This plot shows the real and imaginary parts. $\rho_1 = 2.2 \times 10^3$ [kg/m³], $\rho_2 = 2.5 \times 10^3$ [kg/m³], $\beta_1 = 3$ [km/s], $\beta_2 = 5$ [km/s].

6.2.2 Reflection, refraction, and conversion of P-SV at an internal boundary

Because the derivation is somewhat complex, we only show the final results here^{note 1)}:

$$a = \rho_2 - 2(\mu_2 - \mu_1)p^2 \quad K = a\xi_1 + b\xi_2 \quad N = a\eta_1 + b\eta_2 \quad (6.12)$$

$$b = \rho_1 + 2(\mu_2 - \mu_1)p^2 \quad L = d - 2(\mu_2 - \mu_1)\xi_1\eta_2 \quad M = d - 2(\mu_2 - \mu_1)\xi_2\eta_1 \quad (6.13)$$

$$d = \rho_2 - \rho_1 - 2(\mu_2 - \mu_1)p^2 \quad D = KN + p^2LM \quad (6.14)$$

SV wave incidence

$$R_{SS} = \frac{1}{\Delta} \{ -(a\eta_1 - b\eta_2)K + p^2[d + 2(\mu_2 - \mu_1)\xi_2\eta_1]L \} \quad (6.15)$$

$$R_{SP} = -\frac{\beta_1}{\alpha_1} \frac{2p\eta_1}{\Delta} [ad + 2(\mu_2 - \mu_1)b\xi_2\eta_2] \quad (6.16)$$

$$T_{SS} = \frac{\beta_1}{\beta_2} \frac{2\rho_1\eta_1K}{\Delta} \quad (6.17)$$

$$T_{SP} = \frac{\beta_1}{\beta_2} \frac{2\rho_1p\eta_1L}{\Delta} \quad (6.18)$$

Here,

$$p = \frac{\sin \theta_1}{\alpha_1} = \frac{\sin \theta_2}{\alpha_2} = \frac{\sin \varphi_1}{\beta_1} = \frac{\sin \varphi_2}{\beta_2} \quad (6.19)$$

$$\xi_i^2 = \frac{1}{\alpha_i^2} - p^2, \eta_i^2 = \frac{1}{\beta_i^2} - p^2, \gamma_i^2 = 2\beta_i^2 p^2. \quad (6.20)$$

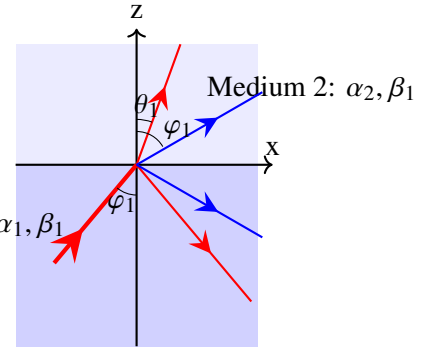


Figure 6.4:

P wave incidence

$$R_{PP} = \frac{1}{\Delta} \{ (a\xi_1 - b\xi_2)N - p^2[d + 2(\mu_2 - \mu_1)\xi_1\eta_2]M \} \quad (6.21)$$

$$R_{PS} = -\frac{\alpha_1}{\beta_1} \frac{2p\xi_1}{\Delta} [ad + 2(\mu_2 - \mu_1)b\xi_2\eta_2] \quad (6.22)$$

$$T_{PP} = \frac{\alpha_1}{\alpha_2} \frac{2\rho_1\xi_1N}{\Delta} \quad (6.23)$$

$$T_{PS} = \frac{\alpha_1}{\alpha_2} \frac{2\rho_1p\xi_1M}{\Delta} \quad (6.24)$$

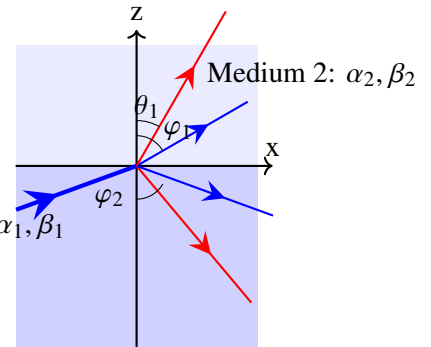


Figure 6.5:

^{note 1)}For a more detailed derivation, please refer to Aki and Richards (2002) or Saito (2009).

$$R_{PP} = \frac{1}{\Delta} \{ (a\xi_1 - b\xi_2)N - p^2 [d + 2(\mu_2 - \mu_1)\xi_1\eta_2]M \} \quad (6.25)$$

$$R_{PS} = -\frac{\alpha_1}{\beta_1} \frac{2p\xi_1}{\Delta} [ad + 2(\mu_2 - \mu_1)b\xi_2\eta_2] \quad (6.26)$$

$$T_{PP} = \frac{\alpha_1}{\alpha_2} \frac{2\rho_1\xi_1N}{\Delta} \quad (6.27)$$

$$T_{PS} = \frac{\alpha_1}{\alpha_2} \frac{2\rho_1p\xi_1M}{\Delta} \quad (6.28)$$

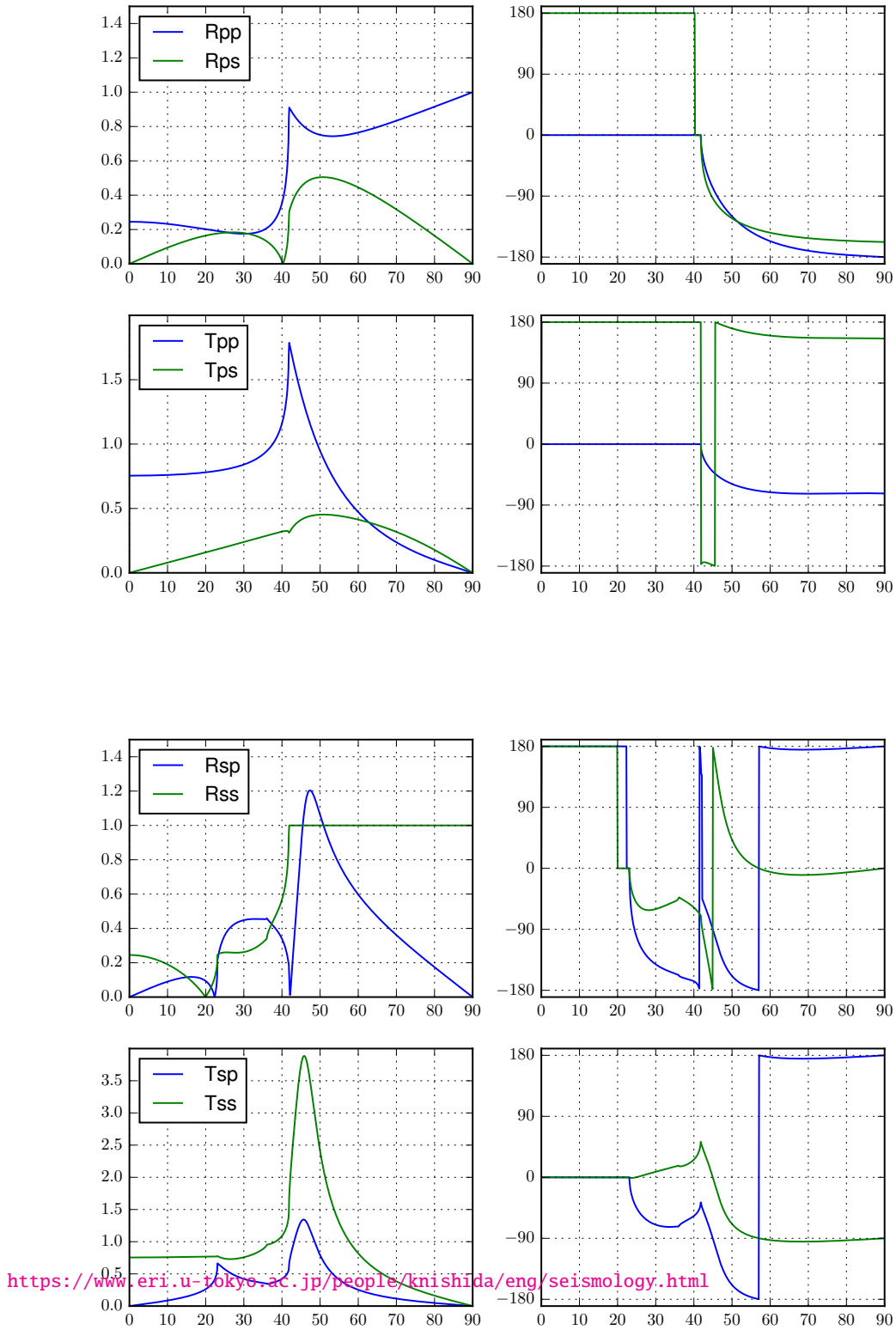


Figure 6.6: Reflection, transmission and conversion coefficients at a boundary taken from Saito (2016).⁽⁶⁾

6.2.3 Asymptotics for near vertical incident

This section describes the asymptotics of the coefficients for the near-vertical incident (p is enough small). Here we neglect higher-order terms of p than 2.

P wave incident

$$R_{PP} = \frac{\rho_2 \alpha_2 - \rho_1 \alpha_1}{\rho_2 \alpha_2 + \rho_1 \alpha_1} \quad (6.29)$$

$$R_{PS} = -\frac{2\alpha_1 p [\rho_2 (\rho_2 - \rho_1) \alpha_2 \beta_2 + 2\rho_1 (\mu_2 - \mu_1)]}{(\rho_2 \alpha_2 + \rho_1 \alpha_1)(\rho_2 \beta_2 + \rho_1 \beta_1)} \quad (6.30)$$

$$T_{PP} = \frac{2\rho_1 \alpha_1}{\rho_2 \alpha_2 + \rho_1 \alpha_1} \quad (6.31)$$

$$T_{PS} = \frac{2\rho_1 \alpha_1 p [(\rho_2 - \rho_1) \alpha_2 \beta_1 - 2(\mu_2 - \mu_1)]}{(\rho_2 \alpha_2 + \rho_1 \alpha_1)(\rho_2 \beta_2 + \rho_1 \beta_1)}. \quad (6.32)$$

S wave incident

$$R_{SS} = \frac{\rho_2 \beta_2 - \rho_1 \beta_1}{\rho_2 \beta_2 + \rho_1 \beta_1} \quad (6.33)$$

$$R_{SP} = -\frac{2\beta_1 p [\rho_2 (\rho_2 - \rho_1) \alpha_2 \beta_2 + 2\rho_1 (\mu_2 - \mu_1)]}{(\rho_2 \alpha_2 + \rho_1 \alpha_1)(\rho_2 \beta_2 + \rho_1 \beta_1)} \quad (6.34)$$

$$T_{SS} = \frac{2\rho_1 \beta_1}{\rho_2 \beta_2 + \rho_1 \beta_1} \quad (6.35)$$

$$T_{SP} = \frac{2\rho_1 \beta_1 p [(\rho_2 - \rho_1) \alpha_2 \beta_1 - 2(\mu_2 - \mu_1)]}{(\rho_2 \alpha_2 + \rho_1 \alpha_1)(\rho_2 \beta_2 + \rho_1 \beta_1)}. \quad (6.36)$$

Here, R_{PP} and R_{SS} can be represented by impedance, similarly to SH waves. Surprisingly, the conversion coefficients (e.g., T_{PS} and R_{PS}) are explicitly sensitive to both density contrast and S-wave velocity contrast. Figure 6.6 shows that this first-order approximation remains valid over a large range of slowness values p .

§6.3 Radiation of seismic wave from a point source: wavefront and ray path

here we consider propagations of a wave packet $f(t)$. Amplitude in y component s_y is given by

$$s_y(\mathbf{x}, t) = A(\mathbf{x})f(t - T(\mathbf{x})), \quad (6.37)$$

where $T(\mathbf{x})$ is the arrival time at a location $\mathbf{x}$, and $A(\mathbf{x})$ is the amplitude. The isocontour for the same arrival time of $T(\mathbf{x})$ is known as the "wavefront". The ray is perpendicular to the wavefront. The direction of the ray is represented by $\hat{\mathbf{n}} \equiv \beta \nabla T$, where β is the S-wave velocity. The comparison with a plane wave shows that ∇T corresponds to the slowness vector.

Figure 6.7 shows a typical example of wavefronts for two-layer medium. The wavefronts are associated with ripples spread out when a single pebble is dropped into water.

For understanding ray paths, the physical interpretation is feasible. Here we consider that the spatial variation of A is enough longer than the wavelength. Because the spatial derivative of A is negligible, the displacement is written as,

$$\nabla s_y = -A \nabla f = -A \frac{df}{dt} \nabla T \quad (6.38)$$

The energy flux $\mathbf{K}$ is given by

$$\mathbf{K} = \beta \hat{\mathbf{n}} \rho \dot{f}^2 A^2. \quad (6.39)$$

Conservation of energy along the ray path gives us the information of the amplitude. Figure 6.8 shows a typical example of ray paths.

A comparison of ray paths for reflection before the critical angle with the corresponding wavefronts in Figure 6.7 is easy to interpret. On the other hand, the behavior after the critical angle is complex. Total reflection after the critical angle causes the split between the transited and reflected wave packets. The following section explains the behavior.

§6.4 Wave behavior at a seismic discontinuity

In the previous chapter, we learned about the reflection and transmission of plane waves at a material boundary. Building on those results, let us consider the physical behavior of seismic waves excited by a point source (i.e., a Green's function) located above a discontinuity. The resulting wave modes are generally categorized into four types: direct waves, reflected waves, transmitted waves, and head waves.

6.4.1 Direct wave

First, let us consider the direct wave. For simplicity, we neglect free surfaces on the ground. The green line in Figure 6.8 shows the direct wave. The SH-wave propagation can be represented by a Green's function in an infinite homogeneous medium. Of course, the travel time T is proportional to the epicentral distance between the source and the station as

$$T(r) = r/\beta_1, \quad (6.40)$$

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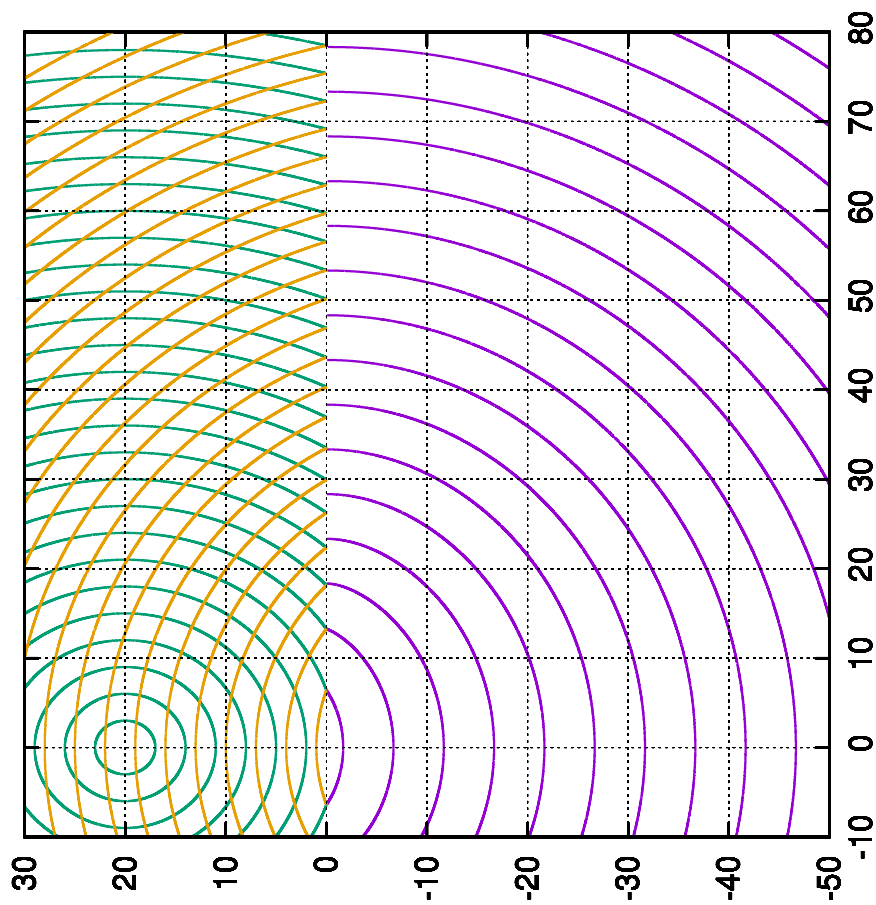


Figure 6.7:

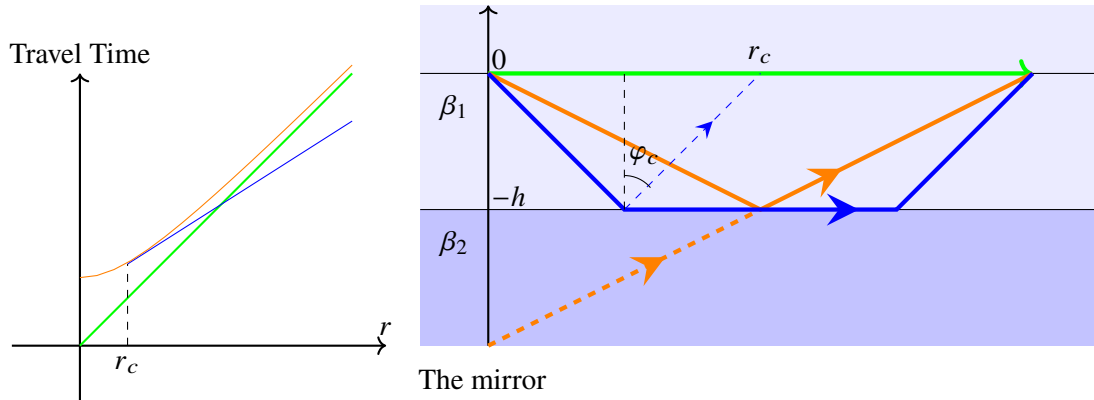


Figure 6.8:

where r is the epicentral distance for the station and source in the xy plane. For a point source in the 3-D medium, the amplitude decreases with $1/r$, which is given by the conservation of energy along the path as,

$$A_d \sim \frac{1}{4\pi r} e^{ik_{\beta_1} r \mu_1 \beta_1} \quad (6.41)$$

6.4.2 Reflected wave

Next, let us consider the reflected wave. The orange line shows the reflected wave. The mirror-symmetric source for the discontinuity (at $z = -h$) helps us intuitively understand its ray path. The travel time is given by:

$$T = 2\sqrt{h^2 + (r/2)^2}/\beta_1, \quad (6.42)$$

and the amplitude is given by:

$$A_r \sim \frac{1}{4\pi R_0 \mu_1 \beta_1} R_{12}(\mathbf{p}) e^{ik_{\alpha_1} R_0}, \quad (6.43)$$

where R_{12} is the reflection coefficient. Because R_{12} is real before the critical angle, the reflection does not induce a phase shift. However, total reflection beyond the critical angle causes a phase shift, and its amplitude becomes 1. These are also known as post-critical reflections or wide-angle reflections.

Beyond the critical angle, the behavior becomes more complex, featuring the splitting of wavefronts and significant phase shifts. The properties of head waves are key to understanding this complexity, as will be explained in the next subsection.

6.4.3 Transmitted wave

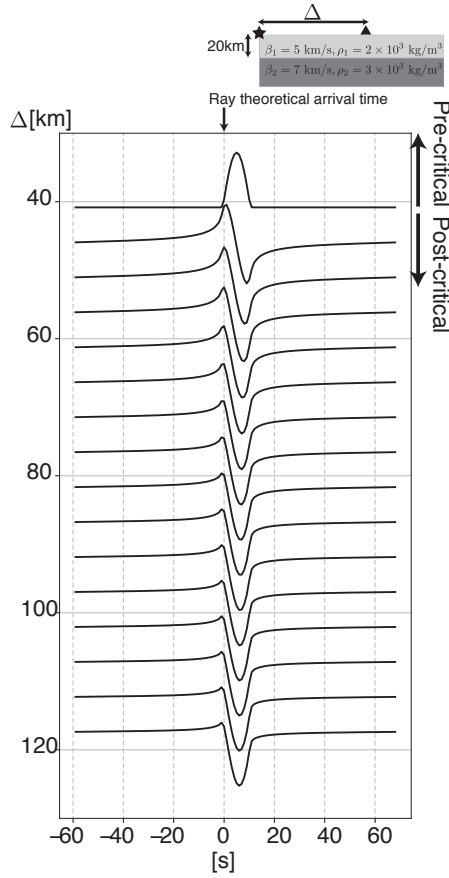


Figure 6.9: Example of wide-angle reflection. After the critical angle, we can see the reflected wave arrives before the ray theoretical value because of the phase shift due to the inhomogeneous wave.

Last, let us consider the transmitted wave. Because this wave does not return to the surface, of course, we cannot measure the travel time at a surface station.

$$A_t \sim \frac{1}{4\pi\sqrt{r}} T_{12}(\mathbf{p}) \left(r_1 + r_2 \frac{\beta_2 \cos^2 \varphi_1}{\beta_1 \cos^2 \varphi_2} \right)^{-1/2} e^{i(k_{a1}r_1 + k_{a2}r_2)} \quad (6.44)$$

where r_1 is the length of segment OA, r_2 is the length of segment AB, and r is the length of segment OB. A wavefront interacts with the discontinuity at an incident angle φ_1 , and it is transmitted at an emergence angle φ_2 . Let us estimate the amplitude at point B based on energy conservation. Figure 6.10 illustrates a cross-section of the 3-D medium along the line OB. Here, δl represents the interception

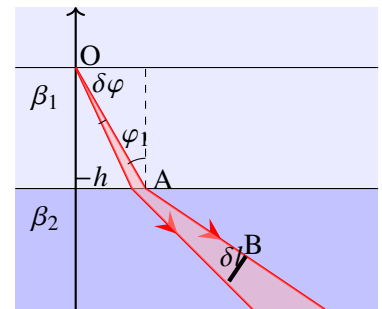


Figure 6.10:

length at point B, perpendicular to the ray path. A straightforward geometrical estimation yields:

$$\delta l = \frac{\cos \varphi_2}{\cos \varphi_1} \left(r_1 + r_2 \frac{\beta_2 \cos^2 \varphi_1}{\beta_1 \cos^2 \varphi_2} \right) \delta \varphi. \quad (6.45)$$

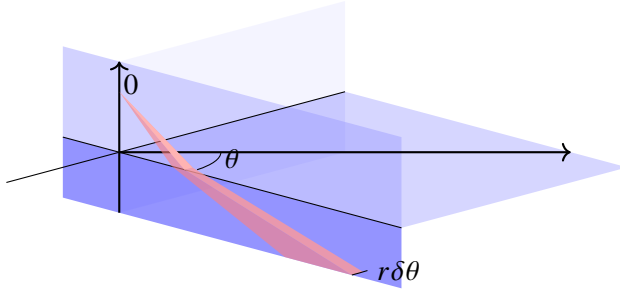


Figure 6.11: Enter of a ray path toward the discontinuity. The ray path is refracted to the radial direction, whereas it is not toward the tangential direction.

Figure 6.11 shows a bird's view of the ray path. The ray path is refracted to the radial direction (on rz plane), whereas it is not toward the tangential direction (θ). Therefore, a cross-section area at point r_0 away from the origin is given by $\delta r_0^2 \delta \theta \delta \varphi$, and a cross-section area at point B is given by $r \delta \theta \delta l$. Here we define amplitude A_i on a unit sphere in medium 1. The conservation of energy along the ray leads to

$$\left(\frac{\rho_2 \beta_2 \cos \varphi_2}{\rho_1 \beta_1 \cos \varphi_1} |T_{12}|^2 \right) \rho_1 A_1^2 \omega^2 \beta_1 \delta \varphi r_0^2 \delta \theta = \rho_2 A_t^2 \omega^2 \beta_2 \delta l r \delta \theta. \quad (6.46)$$

Note that the transmission coefficient is multiplied by $\frac{\rho_2 \beta_2 \cos \varphi_2}{\rho_1 \beta_1 \cos \varphi_1}$.

Energy normalization of reflection and transmission coefficients

In the previous chapter, we introduced the reflection and transmission coefficients. They are traditionally defined by the amplitude ratio between the scattered wave and the incident wave. However, when considering energy conservation, an energy-normalized definition of these coefficients makes their physical meaning more explicit.

Conservation of energy (Eq. 6.47) for the reflection and transmission is given by

$$\rho_1 \beta_1 \cos \varphi_1 = \rho_1 \beta_1 \cos \varphi_1 |R_{12}|^2 + \rho_2 \beta_2 \cos \varphi_2 |T_{12}|^2. \quad (6.47)$$

Both sides divided by $\rho_1 \beta_1 \cos \varphi_1$ leads to

$$1 = |R_{12}|^2 + \frac{\rho_2 \beta_2 \cos \varphi_2}{\rho_1 \beta_1 \cos \varphi_1} |T_{12}|^2. \quad (6.48)$$

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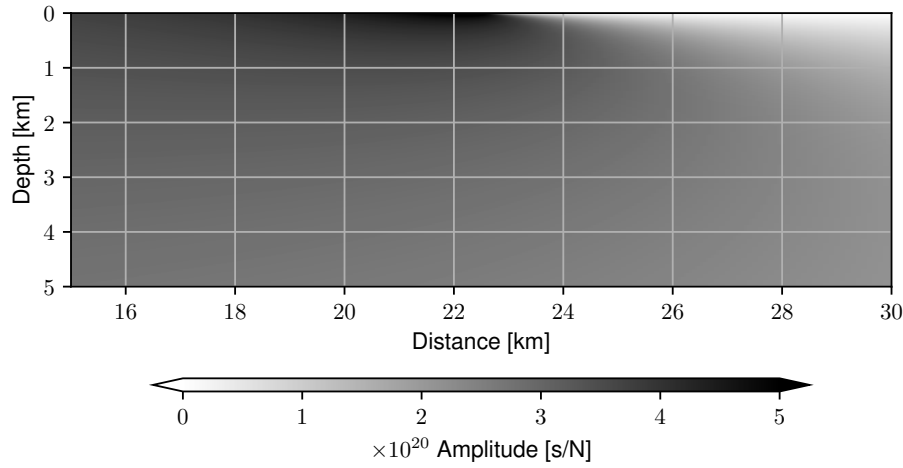


Figure 6.12: Spatial distribution of the Fourier amplitudes of the transmitted wave. The direct wave with the critical angle enters at the point of 22.5 km. $\rho_1 = 2$ [kg/m²], $\rho_2 = 2.2$ [kg/m²], $\beta_1 = 3.0$ [km/s], $\beta_2 = 5$ [km/s].

Here we define a energy normalized transmission coefficient T_{12}^{norm} as,

$$T_{12}^{norm} = \sqrt{\frac{\rho_2 \beta_2 \cos \varphi_2}{\rho_1 \beta_1 \cos \varphi_1}} T_{12}^{\text{note 2)}. \quad (6.49)$$

The behavior near the critical angle

Let us consider the behavior of the transmission wave near the critical angle.

First, let us examine the amplitude of the transmitted wave. Figure 6.12 shows its amplitude as a function of the horizontal position and depth below the boundary. The critical angle corresponds to a horizontal distance of 22.5 km. Beyond the critical angle, the amplitude of the transmitted wave diminishes sharply. This sudden drop in amplitude can be understood as a diffraction phenomenon occurring in that region.

^{note 2)} See a textbook by Shearer⁽⁸⁾ for details. Sections of ray theory are easy-to-grasp.

Next, we consider the behavior of the transmission wave amplitude near the critical angle. Before the critical angle, because $r_2 = 0$ at the surface, the amplitude of the transmitted wave is given by

$$A_t \sim \frac{1}{4\pi\sqrt{r}\mu_1\beta_1} T_{12}(\mathbf{p}) r_1^{-1/2}. \quad (6.50)$$

After the critical angle, the finite length of r_2 and $\cos \varphi_2 = 0$ lead to $A_t \sim 0$ (Figure 6.13). Without

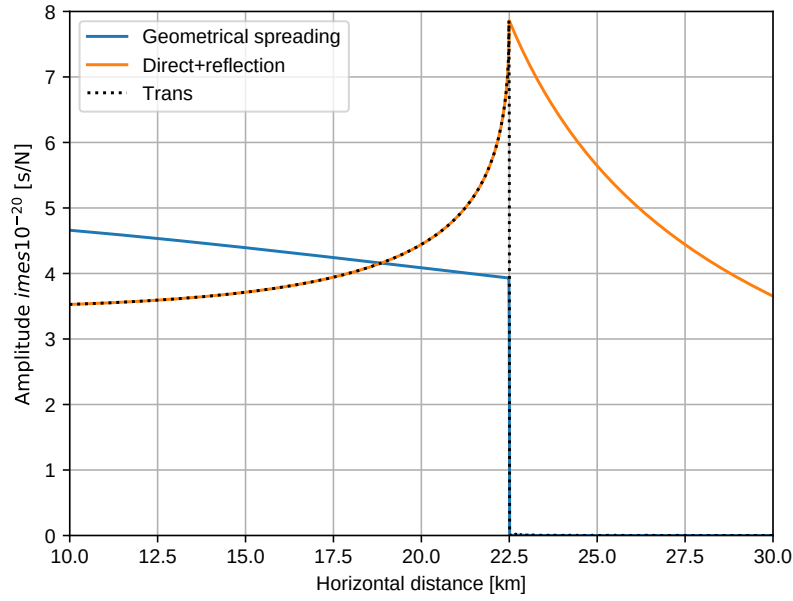


Figure 6.13: Amplitudes of transmitted wave (black) and the direct + reflection (red). The blue line shows the transmitted wave amplitude without the transmission coefficient. $\rho_1 = 2$ [kg/m²], $\rho_2 = 2.2$ [kg/m²], $\beta_1 = 3.0$ [km/s], $\beta_2 = 5$ [km/s].

considering T_{12} , A_t is proportional to $1/r$ approximately. The amplitude changes with distance gently (Figure 6.13). However, the transmission coefficient changes drastically near the critical angle. Accordingly, A_t has a peak near the critical angle ($r_c/2 = 22.5$ km).

Next, let us evaluate this sharpness. Let l be the horizontal distance from $r_c/2$. Consider the case where the incident angle is smaller than the critical angle by a small angle $\delta\varphi_2$. If we set the refraction angle $\varphi_2 = \pi/2 - \delta\varphi_2$, from Snell's law, we have

$$\frac{\tan \varphi_2 \delta\varphi_2^2}{2} \sim \delta\varphi_1, \quad (6.51)$$

This relationship is obtained. Additionally, from geometric properties,

$$l \sim \frac{h}{\cos^2 \varphi_c} \delta\varphi_1, \quad (6.52)$$

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This relationship is obtained. Using these relational expressions,

$$T_{12} \sim \frac{2}{1 + \frac{\rho_2 \beta_2}{\rho_1 \beta_1} \sqrt{\frac{2l \cos \varphi_c}{h \sin \varphi_c}}} \quad (6.53)$$

Using these approximations, the characteristic width of the peak, defining the spatial extent of the diffraction effect l_d , becomes:

$$l_d \sim \frac{(\rho_1 \beta_1)^2 \sin \varphi_c}{(\rho_2 \beta_2)^2 \cos \varphi_c} \frac{h}{2} = \frac{\rho_1^2 \beta_1^3}{\rho_2^2 \beta_2^3 \cos \varphi_c} \frac{h}{2}. \quad (6.54)$$

We can expect the amplitude of the head wave to scale proportionally with the length l_d .

6.4.4 Head wave

The third type is a head wave (also known as a re-fracted wave). The wave enters the second layer with a critical angle, and it propagates in the horizontal direction along the uppermost part of the second layer (Figure 6.8). The travel time T_H is given by

$$T_H(r) = \frac{r - r_c}{\beta_2} + \frac{r_c}{\cos \varphi_c \beta_1}. \quad (6.55)$$

The amplitude A is written by

$$A_{head} \sim \frac{i}{2\pi\omega\mu_1\beta_1} \frac{\rho_1\beta_1^2}{\rho_2\beta_2(1 - \beta_1^2/\beta_2^2)} \frac{1}{\sqrt{r}D^{3/2}} e^{i\omega t_h} \quad (6.56)$$

Although the frequency- and distance- dependencies are difficult to understand, I try to explain in an intuitive manner^{note 3)}.

Figure 6.7 shows wavefronts of the direct and the reflected and refracted wave share a point on the boundary before reaching the critical angle. After the critical reflection, they are split into two groups: one is a direct and reflected wave, and the other is a head wave and transmitted wave. Exactly speaking, inhomogeneous waves along the boundary also exists.

Let us consider the details of the split. A major difference between the two groups originated from the fact that the direct wave cannot enter the second layer into the second medium. When the transition from the transmitted wave to the inhomogeneous wave in the second layer, let us consider a secondary point source with a spatial scale of about the wavelength at r_c (finite frequency effect) based on the representation theorem, which is a natural extension of Huygens principle. As in the

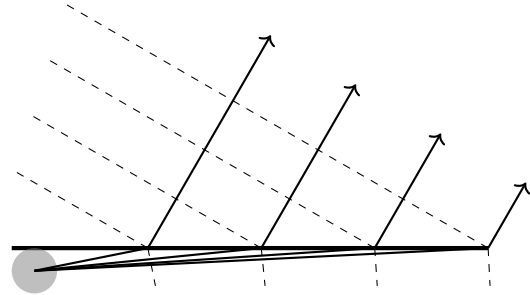


Figure 6.14: A schematic figure of head wave based on Huygens' principle.

^{note 3)} Looking at Eq. 5.92, we see that $k_z = 0$ is a singular point. The singular point is the region where $k = \omega/\beta_2$, which corresponds to where the head wave occurs.

case of diffraction at a narrow slit, an SH wave is radiated in the right direction. Although the ray path of the head wave seems to be parallel to the boundary in layer 2, the path is slightly inclined to owe the source depth of an about wavelength. Thus SH wave is transmitted toward layer 1, as predicted by the Huygens principle. The incident angle can be estimated to be $90^\circ \varphi_1 \sim D/\lambda$. The corresponding transmission coefficient T_{12} can be approximated by

$$T_{21} = \frac{2\rho_2\beta_2 \cos \varphi_2}{\rho_1\beta_1 \cos \varphi_1 + \rho_2\beta_2 \cos \varphi_2} \sim \frac{2\rho_2\beta_2\lambda/D}{\rho_1\beta_1 \cos \varphi_c} = \frac{4\pi\rho_2\beta_2^2}{\rho_1\beta_1} \frac{1}{D\omega \cos \varphi_c} \quad (6.57)$$

The amplitude decreases as $r^{-1/2}D^{-1}D^{-1/2}$, where (1) $r^{-1/2}$ represents geometrical spreading of the wavefront in xy plane, (2) $D^{-1/2}$ represents geometrical spreading of the wavefront in xz plane, and (3) D^{-1} represents contribution by the transmission coefficient. The frequency dependence of ω can be explained by the frequency dependence of the incident angle φ_1 , which is proportional to the inverse of the wavelength. This is why the low-frequency component of the head wave is emphasized.

The phase is shifted by 90 degrees. This is because the phase shifts when the wavepacket splits. Because we are considering a 3-D problem, the split of the wavepacket occurs in a ring shape. For this reason, we can understand it by analogy to the 2-D Green's function, where the phase shifts by $\pi/4$ at the origin (hence the Bessel function appeared). When the wave diverges, the phase shifts by $\pi/4$, but this time both convergence and divergence happen, so the phase shifts by $2 \times \pi/4$, resulting in a $\pi/2$ phase shift^{note 4)}. This equation exhibits that the head wave can be represented by the integration of the direct wave in time domain^{note 5)}. An example of observation in Figure 6.19 shows the dominance of low-frequency components in the head wave.

By considering the head wave as a type of diffracted wave, its amplitude can be estimated. As evaluated in Eq. 6.54, let us assume that the head wave is a wave leaking from a length l_d . In this case, the amplitude should be proportional to l_d , and thus the amplitude should be proportional to $T_{12}l_d$. In this case,

$$A_t \propto \frac{\rho_1\beta_1^2}{\rho_2\beta_2 \cos^2 \varphi_c} = \frac{\rho_1\beta_1^2}{\rho_2\beta_2(1 - \beta_1^2/\beta_2^2)}. \quad (6.58)$$

This dependence explains the amplitude dependence on density and seismic velocity.

Because a realistic Earth structure is more complex, the ray path is not so simple. For example, the Pn wave (head wave for Moho discontinuity) is refracted in the uppermost mantle: the behavior may be easy to understand. The next section explains such a case.^{note 6)}

Thus, this section explains the behaviors of Green's function for a two-layer medium. In the next section, I will introduce ray theory for discussions of seismic wave propagations in a multi-layer structure.

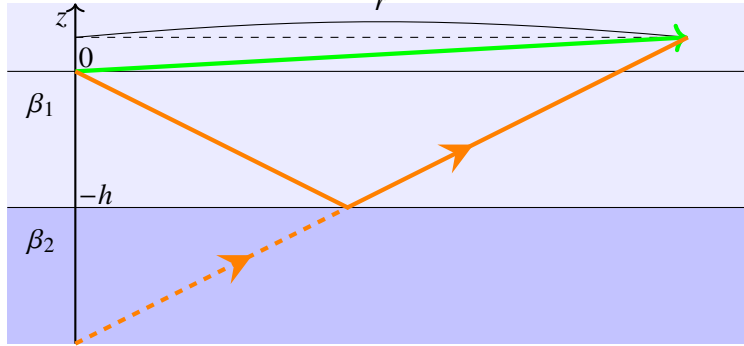
^{note 4)}This is strongly related to the Caustic explained in Section ???. Intuitively, when a wave converges or diverges at a certain point, it becomes a singular point. Considering the connection of solutions, it can be interpreted that the coordinate system flips.

^{note 5)}A more exact discussion mathematically is given by Aki and Richards⁽¹⁾ or Saito⁽⁶⁾ (Chapters 8, 10). In the textbooks, the complex integrals are evaluated. On the other hand, we sacrifice some mathematical rigor to explain the shape of the equations with physical intuition in this chapter.

^{note 6)}See Stein and Wyession⁽⁹⁾ §3.2.3 for details.

Problem 6.1

Derive Eq. 6.45. This is a simple geometrical problem.



The mirror

Figure 6.15:

6.4.5 Evaluation of integral: stationary phase approximation

The previous sub-section physically interpreted direct, reflected, transmitted, and head waves. Now let us consider them more quantitatively using the stationary phase method.

As mentioned at the beginning of this chapter, wave propagation in a two-layer medium can be exactly evaluated by decomposing it into plane waves and multiplying each plane wave by the reflection-transmission coefficient and integrating it in the wavenumber domain. This expression can be naturally extended to cylindrical waves (Hankel function) and is expressed as The potential χ_d representing a direct wave is given by the integration of slowness as

$$\chi_d = \frac{\omega}{4\pi\mu_1} \int_{-\infty}^{\infty} H_0^{(1)}(\omega pr) \frac{e^{-i\omega\xi_1|z|}}{-2i\xi_1} p dp. \quad (6.59)$$

Here, let us consider a case of ξ_1 becomes imaginary. As χ_d is physically meaningful (not diverging) for $p \rightarrow \infty$, we define the sign as

$$\xi_1 = \begin{cases} \sqrt{\beta_1^{-2} - p^2} & |p| \leq 1/\beta_1, \\ \sqrt{p^2 - \beta_1^{-2}}i & |p| > 1/\beta_1. \end{cases} \quad (6.60)$$

Potentials χ_r and χ_t for reflected and transmitted waves are analogously given by:

$$\chi_r = \frac{\omega}{4\pi\mu_1} \int_{-\infty}^{\infty} R_{12}(p) H_0^{(1)}(\omega pr) \frac{e^{i\omega\xi_1|z+2h|}}{-2i\xi_1} p dp, \quad (6.61)$$

$$\chi_t = \frac{\omega}{4\pi\mu_1} \int_{-\infty}^{\infty} T_{12}(p) H_0^{(1)}(\omega pr) \frac{e^{i\omega(\xi_1 h - \xi_2(z+h))}}{-2i\xi_1} p dp. \quad (6.62)$$

It should be noted that the reflection potential χ_r contains the contributions of both the reflected and head waves.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

This integral is numerically evaluated in Figure 6.16. ^{note 7)} The direct wave is impulsive, showing that it propagates without changing its shape. The head wave is clearly visible from around 60 km, and it is also evident that it is predominantly long-period. The reflected waves are pulse-like at stations close to the epicenter (up to about 20 km), but when it exceeds the critical angle (wide-angle reflection), the reflected waves gradually shift in phase.

The numerical integration results show that, indeed, this integral (Weyl's table expression) seems to be correct. To grasp this integral roughly, let us consider its approximate behavior. Here we will consider direct-directed wave as the simplest case.

First, assuming that the propagation distance is sufficiently large compared to the wavelength, the Hankel function is approximated by

$$H_0^{(1)}(z) \sim \sqrt{\frac{2}{\pi z}} e^{i(z - \pi/4)}. \quad (6.63)$$

Then, χ_d can be approximated by

$$\chi_d \sim \frac{1}{4\pi\mu_1} \sqrt{\frac{2\omega}{\pi r}} e^{-i\pi/4} \int_{-\infty}^{\infty} \frac{e^{i\omega(pr - \xi_1|z|)}}{-2i\xi_1} \sqrt{p} dp. \quad (6.64)$$

Here we consider a case for $z=20$ km, $r = 30$ km, $f = 2\pi 5$ [Hz], and $\beta = 3$ km/s, Figure 6.17 plot the integrand

$$\frac{e^{i\omega(pr - \xi_1|z|)}}{-2i\xi_1} \sqrt{p}. \quad (6.65)$$

The figure shows the oscillatory shape in most locations, but the oscillation stops near $p\beta_1 = 0.8$. When integrated, the contribution near this point becomes large.

Let us now evaluate the integral using stationary phase method. Because the part of the gentle oscillation shows a location where the phase change is small,

$$\frac{d\omega(pr - \xi_1|z|)}{dp} = 0. \quad (6.66)$$

Then,

$$\frac{d(pr - \xi_1|z|)}{dp} = r - |z| \frac{d\xi_1}{dp} = r - \frac{|z|p}{\xi_1} = 0. \quad (6.67)$$

Therefore, the contribution of the integral near p (called the stationary point) which satisfy

$$p_0 = \sqrt{\frac{r^2}{\beta_1^2 d^2}} \quad (6.68)$$

becomes large, where d is the distance $\sqrt{r^2 + z^2}$.

^{note 7)} ξ_1 has a singularity. In order to avoid the singularity in the numerical integration, we gave a very small imaginary part to the elastic constants. This physically corresponds to giving a weak damping

Although the calculations are a little complicated ^{note 8)}, the integrand can be evaluated by Taylor expansion up to the second order. Because the first-order term disappears from the condition to be the stationary point, we can evaluate the corresponding phase up to the second order as

$$\omega(pr - \xi_1|z|) = \omega \left(\frac{d}{\beta_1} - \frac{d^3\beta_1}{|z|^2} \frac{(p - p_0)^2}{2} \right). \quad (6.69)$$

With an assumption that ξ_1 and p are small enough near the stationary point, the integrand χ_d can be evaluated up to the second order terms of the Taylor expansion as

$$\chi_d \sim \frac{1}{4\pi\mu_1} \sqrt{\frac{2\omega}{\pi r}} e^{-i\pi/4} \int_{-\infty}^{\infty} \frac{e^{i\omega \left(\frac{d}{\beta_1} - \frac{d^3\beta_1}{|z|^2} \frac{(p - p_0)^2}{2} \right)}}{-2i \frac{|z|}{d\beta_1}} \sqrt{p_0} dp, \quad (6.70)$$

where

$$\xi_1|_{p=p_0} = \frac{|z|}{d\beta_1}. \quad (6.71)$$

This integral can be calculated by using the Fresnel integral

$$\int_{-\infty}^{\infty} e^{-ia^2x^2} dx = \frac{1}{|a|} \sqrt{\pi} e^{-i\pi/4}. \quad (6.72)$$

Although the calculation is complicated, the potential χ_d can be calculated as

$$\chi_d \sim \frac{1}{4\pi\mu_1} \sqrt{\frac{2\omega}{\pi r}} e^{i\left(\omega \frac{d}{\beta_1} + \frac{\pi}{4}\right)} \frac{d\beta_1 \sqrt{p_0}}{2|z|} \int_{-\infty}^{\infty} e^{-i\omega \left(\frac{d^3\beta_1}{2|z|^2} p^2 \right)} dp \quad (6.73)$$

$$= \frac{1}{4\pi\mu_1} \sqrt{\frac{2\omega}{\pi r}} e^{i\left(\omega \frac{d}{\beta_1} + \frac{\pi}{4}\right)} \frac{d\beta_1 \sqrt{p_0}}{2|z|} \int_{-\infty}^{\infty} e^{-i\omega \left(\frac{d^3\beta_1}{2|z|^2} p^2 \right)} dp \quad (6.74)$$

$$= \frac{1}{4\pi\mu_1} \sqrt{\frac{2\omega}{\pi r}} e^{i\left(\omega \frac{d}{\beta_1}\right)} \frac{\sqrt{r\pi}}{2d\sqrt{\omega}} \quad (6.75)$$

$$= \frac{1}{4\pi\mu_1 d} e^{i\omega \frac{d}{\beta_1}}. \quad (6.76)$$

For an external force

$$H(t) \nabla \times (0, 0, \delta(\mathbf{x})), \quad (6.77)$$

we can evaluate displacement A_d from the potential χ_d as

$$A_d = \frac{1}{4\pi\mu_1 \beta_1 d} e^{i\omega \frac{d}{\beta_1}}, \quad (6.78)$$

which is identical to the exact solution by chance.

^{note 8)} This subsection plant to explain stationary phase method. The following is a simple calculation policy, but since the calculations are complicated. You can skip over the evaluations below.

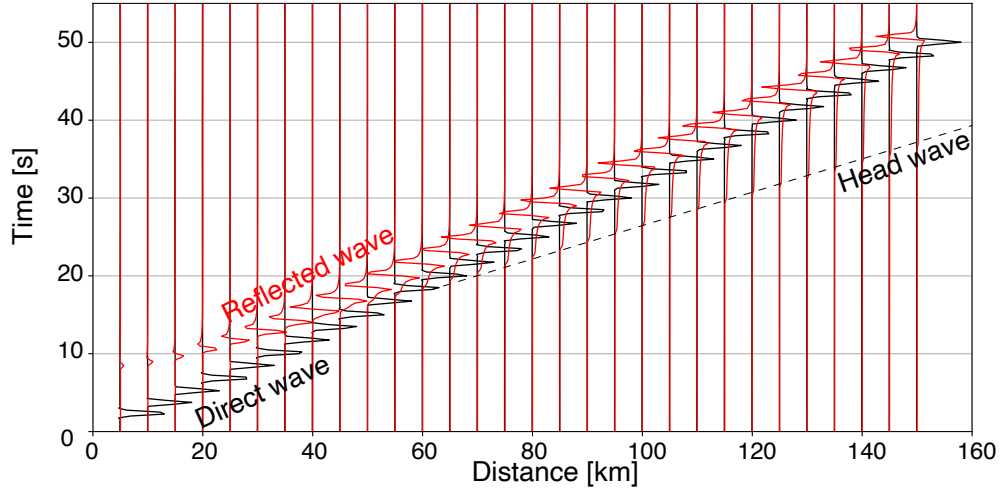


Figure 6.16: An example of numerical evaluation of the integrated for $\beta_1 = 3$ [km/s], $\beta_2 = 5$ km/s and $\rho_1 = \rho_2$. The hypocenter is located at $z = 10$ km and the observed station is located at $z = 15$ km.

Next, let's consider reflected and head waves (χ_r and χ_t). The reflected wave is similar to the direct wave, with a larger contribution from the stationary phase method. Next, let's consider the envelope. You can see that the shape of the envelope changes dramatically at around $p\beta_2 = 1$. This is because the reflection coefficient changes significantly at around the critical angle. At the point, the condition of stationary phase approximation, which is required to apply the stationary phase method, does not hold. Therefore, even when oscillating rapidly outside the stationary point, the integral contribution near $p\beta_1$ does not cancel each other out. This contribution corresponds to the head wave.

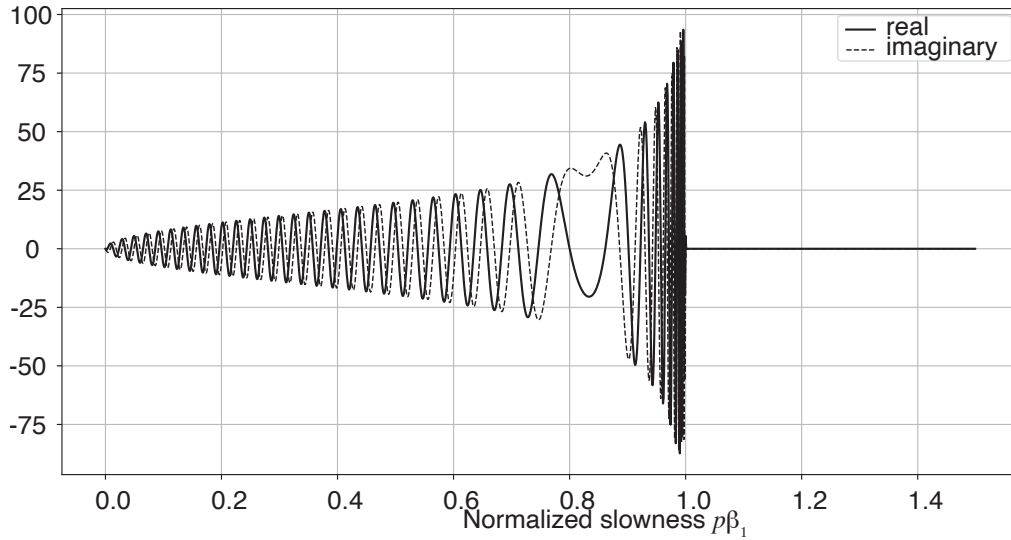


Figure 6.17: The integrand of the direct wave for $z=20$ km, $r = 30$ km, $f = 2\pi 5$ [Hz], $\beta = 3$ km/s. This figure shows rapid oscillations.

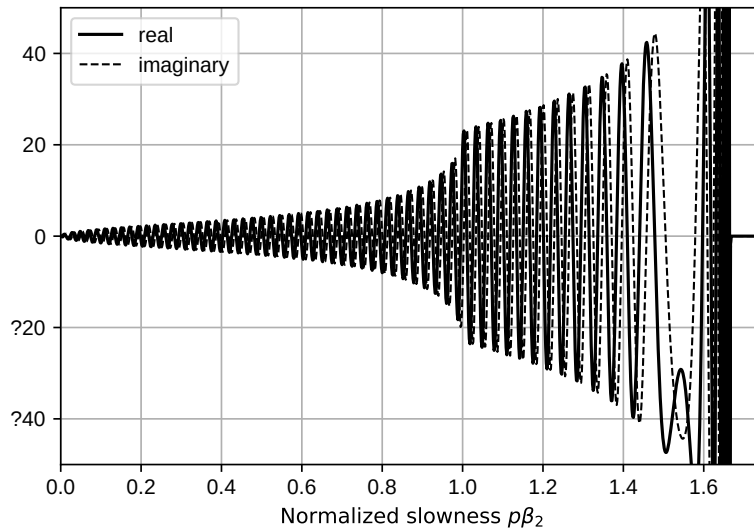


Figure 6.18: The integrand of the reflection wave and the head wave for $z=20$ km, $r = 30$ km, $f = 2\pi 5$ [Hz] and $\beta = 3$ km/s. This figure shows rapid oscillations.

6.4.6 An example of actual records

Figure 6.19 shows a record section when an earthquake with Mw 6.2 at Tottori prefecture on Oct. 21st 2016. The figure shows the first arrivals of Pg (direct wave, which propagates in the crust) up to an epicentral distance of about 170 km. Farther than the distance, the first arrivals are Pn wave arrivals (head wave for Moho discontinuity). The dominant frequency of the Pn wave is longer than the direct wave (Pg wave). In contrast, reflection phases (PmP) exhibit complex wave propagations. The complexity originated from scattering owing to lateral heterogeneities in the crust. At higher frequencies, the travel times of the first arrivals give us robust information because they are not disturbed by the scattering.

The apparent slopes of the Pg and Pn arrivals indicate that $\beta_1 \approx 6$ km/s and $\beta_2 \approx 8$ km/s, respectively. The crossover distance x_d between the Pg and Pn arrivals is given analytically by $x_d = 2h\sqrt{(\beta_1 + \beta_2)/(\beta_2 - \beta_1)}$. Using the observed intersection at roughly 170 km, we can estimate the crustal thickness h to be approximately 30 km. Although this estimation is clearly an oversimplification, a rough calculation like this gives an intuitive grasp of the structural properties. For modeling realistic situations with continuously varying profiles, the ray theory frameworks introduced in the next chapter are much more practical.

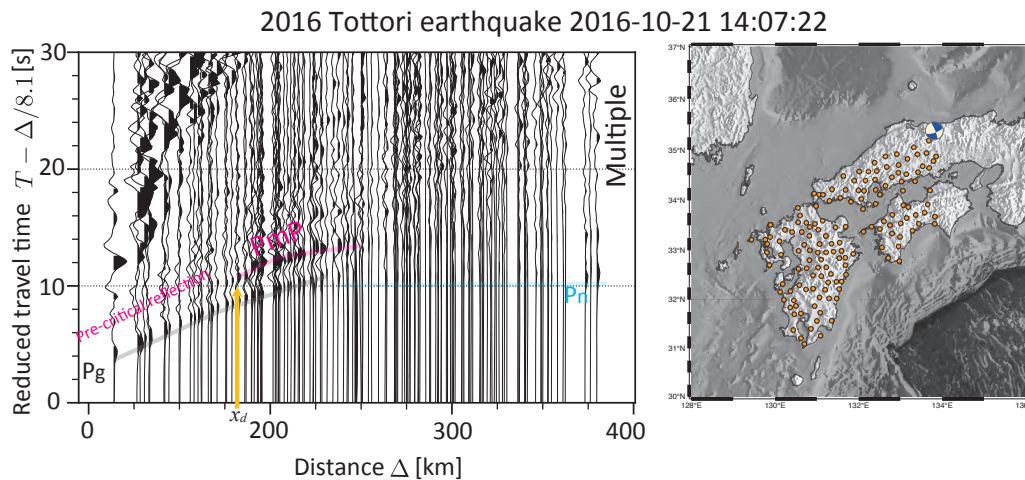


Figure 6.19: Seismograms recorded by Hi-net station against the epicentral distances when an earthquake with Mw 6.2 at Tottori prefecture on Oct. 21st in 2016. Although the first arrivals are easy to pick, the later phases show complex propagation properties. The complex features originated from lateral heterogeneities in the crust, although the lateral heterogeneities in the Chugoku region are weakest in Japan islands

Problem 6.2

The website demonstrates SH-wave propagation in a two-layer medium (2-D). Based on the seismograms, we can estimate (1) the S-wave ratio between the 1st layer and the 2nd layers and (2) the density ratio between the 1st layer and the 2nd layers.

1. Using the whole wave field (depth section in the upper panel), estimate the S-wave ratio between the layer and the density ratio.
2. Using only surface records (seismograms shown in the lower panel) estimate the S-wave ratio between the layer and the density ratio.

To derive an approximation around the critical angle, we can use the expansion detailed in Section 6.4.4 for head waves. Because the travel-time difference $r - T/V$ is smaller, a lower-frequency assumption is necessary, restricting the applicability of the plane wave composition method. Actually, near the antipode of an earthquake, the amplitude of seismic waves becomes exceptionally large due to the focusing of seismic rays. This phenomenon corresponds mathematically to a caustic. This will be explained more thoroughly in the subsequent chapter.

The website demonstrates SH-wave propagation in a two-layer medium (2-D) without the ground (free surface).

1. Run this demo and compare the amplitude of the reflected wave with the results of the reflection coefficient. Pay particular attention to the sign.
2. Discuss the relationship between the head wave and the wide angle reflected wave, which leads near the critical angle.

§6.5 Inhomogeneous wave: Love wave and Scholte wave

In the case of a semi-infinite medium, only a Rayleigh wave can exist as an inhomogeneous wave; what about the two-layer case? Let us consider a scenario where a layer of thickness h is welded to a semi-infinite medium below it (Figure 6.20). An inhomogeneous wave can exist because the wave energy is trapped within the upper, low-velocity layer. First, for the SH-wave case, we will explore a wave known as the Love wave.⁽³⁾ Next, for a situation where the first layer is fluid and the second layer is solid, we encounter the Scholte wave.⁽⁷⁾ We will discuss both types of waves in this section.

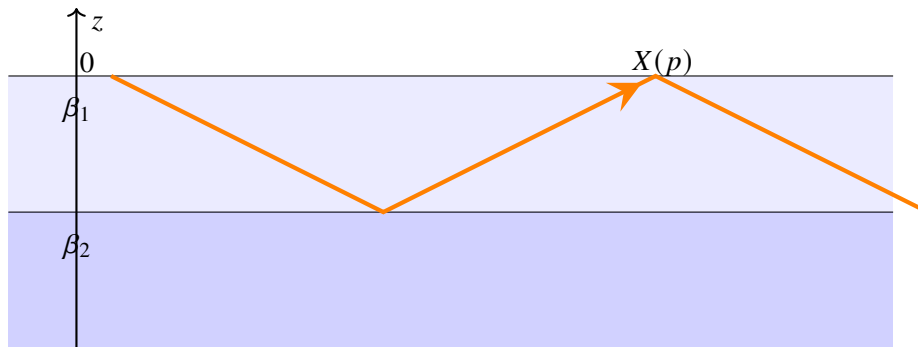


Figure 6.20:

6.5.1 Love wave

First, let us consider how the Love wave propagates (Figure 6.21). $\circ$ in the figure shows a marker for the ground displacement. You can see in the figure how the ground is deformed horizontally. Move the cursor over the figure and press the s key on the keyboard; the Love wave starts to propagate to the right. The circle moving in the front of the page is shown in red and the circle moving in the back of the page is shown in light blue.

Love waves occur when a soft layer overlies a hard layer. For example, a soft crust overlying a stiff mantle. In this figure, circles are placed every 10 km along the vertical axis (depth) and every 6.25 km along the horizontal axis (horizontal). The fourth $\circ$ in the depth direction

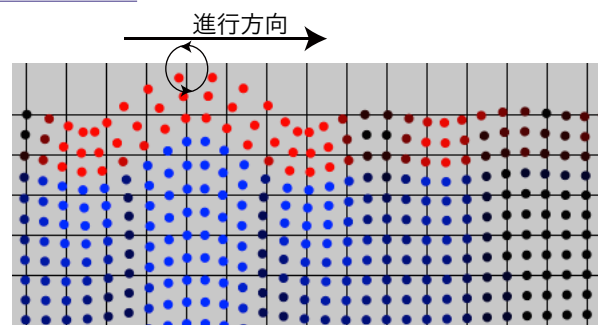


Figure 6.21: A snapshot of a demonstration of Love wave propagation. See the following web application: https://www.eri.u-tokyo.ac.jp/people/knishida/Seismology/Love_wave.html. Play the application for the understandings.

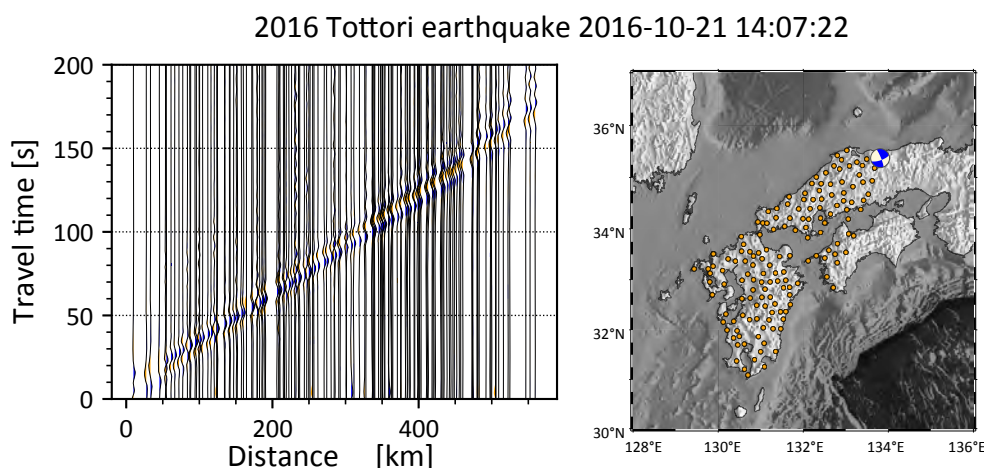


Figure 6.22: Left panel: Example of waveform recorded during the Mj 6.2 earthquake that occurred at a depth of 10 km on October 21, 2016 in Tottori, Japan. The transverse components of Hi-net data are plotted after correcting for the instrumental responses and applying a bandpass filter of 0.02-0.1 Hz. It shows the propagation of Love waves with dispersion. Right panel: The mechanism of the earthquake and the station distribution used in the analysis. Roughly speaking, we chose the stations where the polarities of the Love wave are expected to be the same.

is at the boundary between the crust and the mantle (the Moho discontinuity). You can see how the waves are efficiently propagating horizontally through the crust.

You can see how the shape of the wave has changed compared to the beginning of the wave and its propagation. The red areas extend both horizontally and vertically. On the other hand, the green part is concentrated near the surface. You can see the delayed propagation of the green part. The speed of wave propagation changes depending on the wavelength and the shape of the wave changes. This phenomenon is called dispersion. In this case, waves with longer wavelengths travel faster, and waves with shorter wavelengths travel slower. This is because long waves are strongly affected by the stiff layer (deep layer: mantle).

Next, let us look at the observed waveforms. Figure 6.22 shows the waveforms recorded during the 2016 Tottori earthquake. The wave propagates with dispersion (Love wave propagates with a shape change; see Figure 6.25 for details).

Standing wave

While the previous subsection described the properties of the Love wave qualitatively, physically it can be understood as a standing wave whose energy is trapped in the first layer. Let us consider the extreme case where β_2 approaches infinity (meaning the bottom is a rigid wall). This is analogous to the oscillation of an air column. Assuming the wave propagates vertically, the stress is zero at the free surface, meaning $s_y \propto \cos(kz - \omega t)$. At depth h , the phase shifts by π due to the fixed

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

boundary. However, at the ground surface, the phase does not shift upon reflection because it is an open boundary. Therefore, considering that the phase difference with the wave reflected there is 2π ,

$$k_z \cdot 0 - \omega t = k(2h) - \omega t - \pi - 2n\pi. \quad (6.79)$$

Because $\lambda = 2\pi/k$ and $k_z = k$ in vertical incident case, we obtain the following relation for $n = 0, 1, 2, \dots$:

$$\lambda = \frac{2h}{n + \frac{1}{2}}. \quad (6.80)$$

The wave front reflects off the surface, reflects off the rigid wall, and must shift in phase by 2π before returning to the surface, hence the $n + 1/2$ ^{note 9)}. Let us make a similar estimate for the Love wave below.

Since we are now considering a situation where energy is trapped within the first layer, we assume that the incident wave is totally reflected beyond the critical angle (see Section 6.2.1). The phase of the reflection at the boundary between the first and second layers is shifted by:

$$\arg R_{12} = -2 \tan^{-1} \frac{\mu_2 i p_{2z}}{\mu_1 p_{1z}} = -2 \tan^{-1} \frac{\mu_2 \sqrt{p^2 - \beta_2^{-2}}}{\mu_1 \sqrt{\beta_1^{-2} - p^2}}. \quad (6.81)$$

The difference from the air column case is that once the wave is reflected, it returns to the surface point X (Figure 6.20). This means that the phase of the incident SH wave at $(X, 0)$ must be the same as that of the reflected wave. The two-way travel time T can be written by

$$T = \frac{\sqrt{X^2 + (2h)^2}}{\beta_1}. \quad (6.82)$$

During the propagation, the incident wave advances in phase by Xp . The condition requires the following relation:

$$\omega T + \arg R_{12} = \omega p X + 2n\pi. \quad (6.83)$$

Because

$$X = \frac{2ph}{\sqrt{\beta_1^{-2} - p^2}} \quad (6.84)$$

in summary, we obtain the following relation:

$$\tan \left[h\omega \sqrt{\beta_1^{-2} - p^2} \right] = \frac{\mu_2 \sqrt{p^2 - \beta_2^{-2}}}{\mu_1 \sqrt{\beta_1^{-2} - p^2}}. \quad (6.85)$$

Rearranging this, we obtain Eq. ??, known as the characteristic equation for Love waves. While we adopted a more intuitive explanation here, the formal derivation evaluates the roots of the

^{note 9)} As described in Chapter ??, $T - px$ is a quantity known as the interception time τ . Similar arguments can be made using τ not only for two-layer structures but also for general layered structures.

denominator of the total reflection coefficient (referred to as the secular equation). For further details, please consult Aki and Richards (2002). To improve the perspective of the equation, we can rewrite the relation as

$$\tan\left(\frac{h\omega}{\beta_1}\beta_1\eta_1\right) = \frac{\mu_2}{\mu_1} \frac{\sqrt{1 - (\beta_1/\beta_2)^2 - (\beta_1\eta_1)^2}}{\beta_1\eta_1}, \quad (6.86)$$

where we define $\eta_1 \equiv \sqrt{\beta_1^{-2} - p^2}$, and we choose $\beta_1\eta_1$ as a variable. Figure 6.23 shows the left and right sides of the equation are plotted as functions of $\beta_1\eta_1$, respectively. When the two lines cross each other, the condition is satisfied and a Love wave can exist.

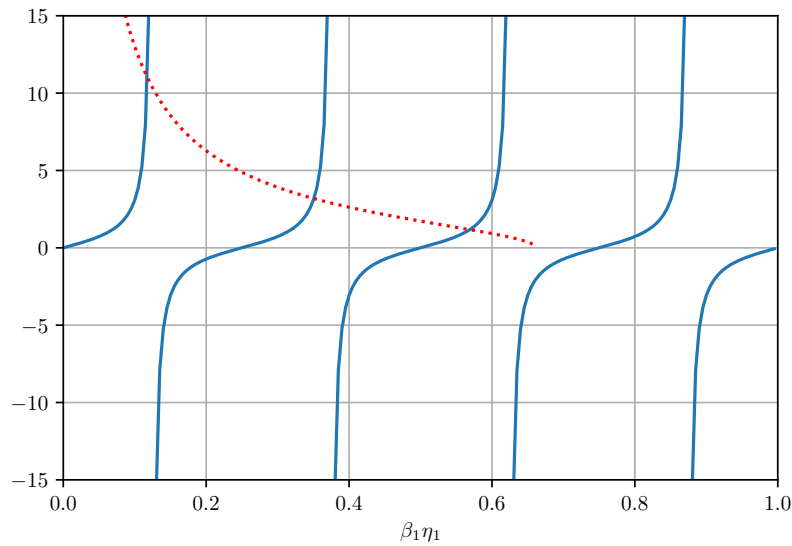


Figure 6.23: Root of the characteristic equation of a Love wave for $h = 30$ km, 0.2 Hz, $\beta_1 = 3$ km/s, $\beta_2 = 4$ km/s, $\rho_1 = 2.5$ g/cm³ and $\rho_2 = 2.8$ g/cm³.

The right side can exist if $\beta_1\eta_1$ is smaller than $\sqrt{1 - (\beta_1/\beta_2)^2}$. Since the left side is $\tan$, we know that at least one solution can exist. This solution is called the fundamental mode. In the current situation (0.2~ Hz), we can see that three solutions exist. They are called the fundamental mode (zeroth-order mode), first higher mode, and second higher mode, respectively, starting from the one with smaller $\beta_1\eta_1$ (i.e., longer wavelength).

Consider the conditions for the existence of a first higher mode. As the frequency decreases, the dashed line moves to the left. It cannot exist when the phase of $\tan$ is left of $3\pi/2$. This requires

$$\pi \frac{\beta_1}{h\omega} \leq \sqrt{1 - (\beta_1/\beta_2)^2}. \quad (6.87)$$

The cut-off frequency of an n th higher mode f_n^{cut} is given by

$$f_n^{cut} = \frac{n\beta_1}{2h} \frac{1}{\sqrt{1 - (\beta_1/\beta_2)^2}}. \quad (6.88)$$

Let us examine the displacement distribution as a function of depth for each mode. Figure 6.24 illustrates the depth distribution of displacement for the different modes. As shown, the energy is confined to the low-velocity layer (layer 1) and decays exponentially with depth in layer 2. Much like the oscillation of an air column, these correspond to the fundamental mode, the first-order mode, and the second-order mode, ordered by the number of nodes.

Phase velocity and group velocity

The phase velocity (p^{-1}) of the Love wave for each frequency is plotted in Figure 6.25. For each mode, the phase speed (solid line) is β_2 on the long-period side and monotonically decreases as the frequency increases, approaching β_1 . The fundamental mode exists from frequency 0, while a higher-order mode has a cutoff frequency on the low-frequency side.

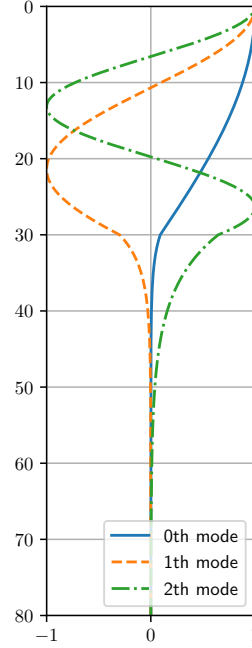


Figure 6.24: Depth distribution of Love wave amplitudes. Amplitudes of each mode are normalized at the surface.

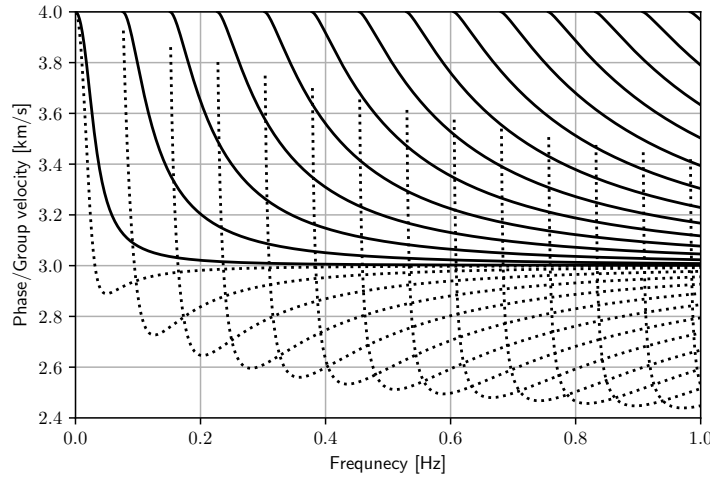


Figure 6.25: Dispersion curves for the Love wave. Phase velocity is shown as a solid line and group velocity as a dashed line. Here we assumed that $h = 30$ km, 0.2 Hz, $\beta_1 = 3$ km/s, $\beta_2 = 4$ km/s, $\rho_1 = 2.5$ g/cm³ and $\rho_2 = 2.8$ g/cm³.

Here is an example of observed dispersion of Love waves. Figure 6.26 shows the dispersion curve of Love waves recorded by Hi-net tiltmeters. It displays the fundamental mode alongside the first, second, and third higher modes, representing the average velocity structure beneath the Japanese Islands. A clear cutoff is visible around 4.5 km/s, indicating that the mantle's S-wave velocity is approximately 4.5 km/s. Furthermore, the phase velocity of the fundamental mode decreases at shorter periods. This happens because the crust cannot be perfectly modeled as a single homogeneous layer; the presence of shallower, slower layers reduces the phase velocity for high-frequency waves.

Surface waves have different propagation speeds at different frequencies. This phenomenon is called dispersion. Here, for simplicity, we consider the following wave propagation:

$$\int_{\omega_0 - \Delta\omega}^{\omega_0 + \Delta\omega} e^{i(k(\omega)x - \omega t)} d\omega, \quad (6.89)$$

where ω is angular frequency, $k(\omega)$ is wavenumber and x represents the location.

The Taylor expansion of $k(\omega)$ around the center frequency ω_0 leads to

$$k(\omega) \approx k_0 + \frac{dk}{d\omega}(\omega - \omega_0) + O(2). \quad (6.90)$$

If $\Delta\omega$ is enough small, we can simplify the equation as

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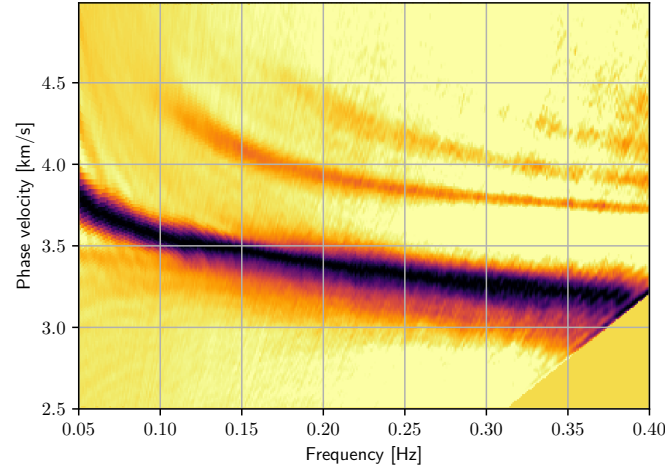


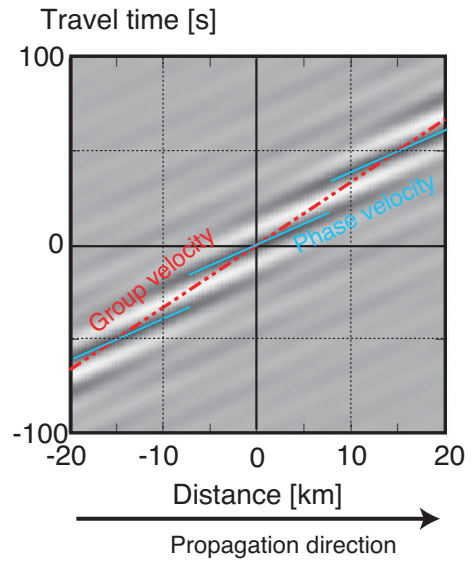
Figure 6.26: Dispersion curves of Love waves propagating through the Japanese Islands observed by Hi-net tiltmeters. See⁽⁵⁾ for details of the analysis.

$$\int_{\omega_0 - \Delta\omega}^{\omega_0 + \Delta\omega} e^{i(k(\omega)x - \omega t)} d\omega \approx e^{i(k_0 x - \omega_0 t)} \int_{-\Delta\omega}^{\Delta\omega} e^{i\omega' \left(\frac{dk}{d\omega} x - t \right)} d\omega' = e^{i(k_0 x - \omega_0 t)} \frac{2 \sin \left(\frac{x}{c_g} - t \right)}{\frac{x}{c_g} - t}, \quad (6.91)$$

where group velocity c_g is defined by $d\omega/dk$. The phase propagates with ω_0/k_0 , and the envelope (sinc function $\sin x/x$) propagates with the group velocity c_g .

The results of this numerical integration are displayed in Figure 6.27. Unlike simpler cases without dispersion, we can see that the waveform changes its shape as it propagates. Curves representing the same phase (or amplitude) propagate at the phase velocity ω/k . Conversely, the wave packet as a whole propagates at the group velocity $d\omega/dk$. By approximating the integral above as a sum of discrete frequencies, the equation for the group velocity can be derived using the sum-to-product trigonometric identities.

Let us return to Figure 6.25, where the group velocity is plotted as a dotted line. While phase velocities decrease monotonically with frequency, group velocities exhibit local minima. What happens near these minima? In those ranges, the group velocity changes very little. If we consider a narrow frequency band (e.g., of width $2\Delta\omega$) around a minimum, wave packets from different



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Figure 6.27: Wave Dispersion.

frequencies within that band travel at nearly the same velocity. Consequently, they arrive simultaneously, resulting in a large combined amplitude. This concentration of energy near a minimum in group velocity is known as the Airy phase^{note 10}. For instance, Figure 6.28 shows an example of the first higher mode of a Love wave. There, it is evident that the wave packets traveling at 2.7 km/s—the minimum group velocity—dominate the seismogram.

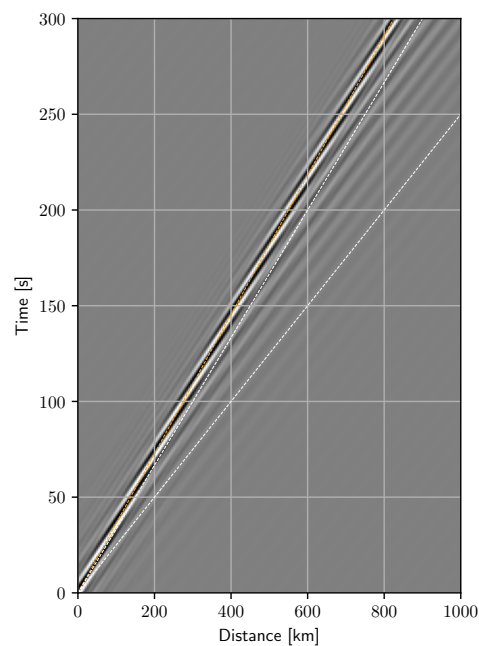


Figure 6.28: First-order higher mode of the Love wave plotted against distance and time. The white dashed lines correspond to phase speeds of 3 km/s and 4 km/s, and the orange dashed line corresponds to 2.7 km/s (the minimum value of group velocity). You can see the dominant wave packet where the group velocity takes a minimum value.

Superposition of Love modes

Finally, let us superimpose all the Love modes. As we first discussed in this section, we can see the wave with wide-angle multiple reflections (at a time shortly before the blue dotted line). The blue dotted line corresponds to the travel time $T(x) = x\beta_2/(\beta_1^2)$ and represents the travel time of the wide-angle multiple reflections. The fact that the wave is not visible at times later than this

^{note 10}For a more detailed treatment, see Saito (2016).

line indicates that it cannot represent a wave incident at a steep angle that would transmit to the layer below. It should be noted that the superposition of modes also reproduces head wave (red dashed line in the figure). While the reflected wave is impulsive, the head wave is bordered. This is because the head wave can be estimated by an integral of the direct waveform (see chapter 6.4.4). This is due to the contribution of higher-order modes near the cutoff frequency. The wave that is parallel to the head wave and visible 20 seconds later is the wave propagating as a head wave after once being reflected wide angle.

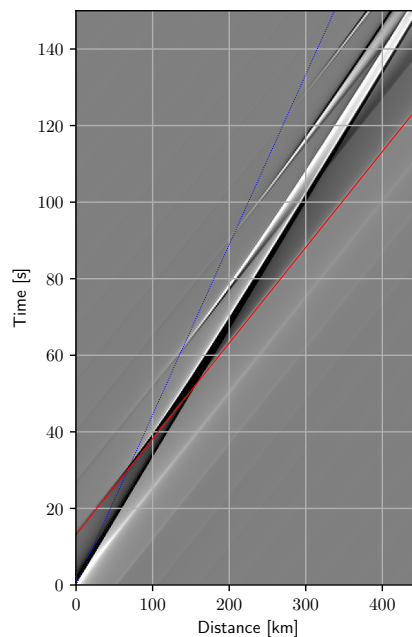


Figure 6.29: The figure shows the superposition of Love modes up to the 14th higher-order modes. The highest frequency is 1 Hz and the structure is the same as in the previous example. The blue dotted line corresponds to the travel time $T(x) = x\beta_2/(\beta_1^2)$ and represents the wide-angle multiple reflection travel time. The red line represents the travel time of the head wave.

One point to note is that we can see wave groups that are clearly physically incorrect (not satisfying the causality) before the head wave. These waves have a phase velocity of approximately 4 km/s (S-wave velocity in the second layer). This is because Love waves cannot represent waves propagating in the second layer. [note 11](#). What waves cannot be represented as a sum of modes can be interpreted physically. More strict treatment will be given in the chapter on normal mode theory.

^{note 11}Strictly speaking, it is the term expressed by the branch integral contribution. See Saito (2009) chapter 9.4

Problem 6.4

Let us evaluate the dispersive wave using the stationary phase approximation.

$$\frac{1}{\pi} \text{Re} \int_0^\infty e^{i(k(\omega)x - \omega t)} d\omega \quad (6.92)$$

Here we define the phase $\Psi = k(\omega)x/t - \omega$, we can rewrite the equation

$$\frac{1}{\pi} \text{Re} \int_0^\infty e^{it\Psi(\omega)} d\omega. \quad (6.93)$$

Because $e^{it\Psi}$ oscillates rapidly, the contribution near

$$\left. \frac{d\Psi(\omega)}{d\omega} \right|_{\omega=\omega_0} = 0 \quad (6.94)$$

became dominant. In this case,

1. Evaluate the integral by expanding ω to the second order term of ω_0 .
2. Consider the group velocity.
3. Also, consider the case where the group velocity takes extreme values.

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